

# From Bragg Rods to Detector Intensity

## Calculation sequence

The calculation follows five physical steps:

1. calculate the continuous scattering intensity along every reciprocal rod;
2. rotate the off-specular rods around the film normal to represent random in-plane orientation;
3. broaden the scattering with the crystallite-tilt distribution;
4. retain the reciprocal-space points that satisfy elastic scattering;
5. project the selected scattering through refraction and detector geometry, then add the weighted incident-beam contributions.

Throughout,  $\alpha$  denotes crystallite tilt,  $\beta$  crystallite azimuth, and  $\phi$  the outgoing-ray azimuth in the caked image. The sections below develop these steps in order and track the intensity normalization at each change of coordinates.

## 1 Finite stacking produces continuous rod intensity

The  $(10L)$  rod is used to introduce the scattering intensity along one reciprocal rod. The same calculation is then applied to every rod allowed by the experimental geometry. For one in-plane Miller pair  $(h, k)$ , the reciprocal vector is

$$\mathbf{Q}_{hk}^0(L) = h\mathbf{a}^* + k\mathbf{b}^* + L\mathbf{c}^*. \quad (1)$$

The coordinate  $L$  is measured in the crystallite frame. The complex scattering amplitude of one quintuple layer in orientation  $\sigma$  is

$$F_\sigma(h, k, L; \lambda) = \sum_a o_a f_a(|\mathbf{Q}|, \lambda) \exp[-W_a(\mathbf{Q})] \exp[i\mathbf{Q} \cdot \mathbf{r}_{a,\sigma}]. \quad (2)$$

Here  $a$  labels an occupied CIF site,  $o_a$  is its occupancy, and  $f_a = f_{0,a}(|\mathbf{Q}|/4\pi) + f'_a(\lambda) + if''_a(\lambda)$  is the wavelength-dependent atomic scattering factor in electron units. The displacement factor is written with  $W_a(\mathbf{Q}) = \frac{1}{2}\mathbf{Q}^\top \mathbf{U}_a \mathbf{Q}$ ; for an isotropic site this becomes  $\frac{1}{2}|\mathbf{Q}|^2 U_{\text{iso},a}$ . The vector  $\mathbf{r}_{a,\sigma}$  is the Cartesian position of site  $a$  in quintuple-layer orientation  $\sigma$ , and the phase convention is  $\exp(+i\mathbf{Q} \cdot \mathbf{r})$ .

In the limit of perfect stacking and infinite coherence, interference between repeated layers places intensity at the allowed reciprocal-lattice positions,

$$I_{hk}^\infty(L) = r_e^2 \sum_{\ell \in \mathcal{A}_{hk}} |F_{hk\ell}|^2 \delta(L - \ell), \quad (3)$$

where  $\mathcal{A}_{hk}$  contains the allowed integer orders. The three-quintuple-layer repeat of the selected  $\text{Bi}_2\text{Se}_3$  stacking sequence gives the strong series  $L = 3n$ .

For a finite ordered stack of  $N$  quintuple layers, the coherent amplitude is

$$A_N^{(0)}(h, k, L) = F_+(h, k, L) \sum_{n=0}^{N-1} \exp\left(\frac{2\pi i n L}{3}\right), \quad (4)$$

and the resulting finite-stack scattering intensity is

$$S_{hk}^{(0,N)}(L) = r_e^2 |F_+(h, k, L)|^2 \left| \frac{\sin(N\pi L/3)}{\sin(\pi L/3)} \right|^2. \quad (5)$$

The quantity  $S_{hk}(L)$  will be called the rod intensity below; in the present normalization it has units of  $\text{\AA}^2$ . At  $L = 3n$ , the interference factor approaches  $N^2$ . A finite stack therefore gives the peaks a width of order  $1/N$  along the crystallographic  $L$  coordinate and produces the finite-size fringes shown in Fig. 1(b).

The stacking model also allows changes in layer registry and orientation. For one possible layer sequence  $\mathcal{P}$ ,

$$A_N(h, k, L; \mathcal{P}) = \sum_{n=0}^{N-1} F_{\sigma_n}(h, k, L; \lambda) \omega_{hk}^{r_n} \exp\left(\frac{2\pi i n L}{3}\right), \quad \omega_{hk} = \exp\left[\frac{2\pi i (h + 2k)}{3}\right]. \quad (6)$$

The probabilities and coherent amplitudes of all allowed layer sequences are combined analytically through the complete stack. The disorder-averaged rod intensity is

$$S_{hk}(L; \lambda) = r_e^2 \left\langle |A_N(h, k, L; \mathcal{P})|^2 \right\rangle_{\mathcal{P}}. \quad (7)$$

This ensemble average is evaluated directly at every crystallographic  $L$  required by the reciprocal-space or detector calculation. The physical progression is

$$\underbrace{\{\ell\}_{\text{Bragg}}}_{\text{infinite ordered crystal}} \longrightarrow \underbrace{S_{hk}^{(0,N)}(L)}_{\text{finite ordered stack}} \longrightarrow \underbrace{S_{hk}(L)}_{\text{stacking model}}. \quad (8)$$

The dimensional distance along the rod is

$$u = |\mathbf{c}^*|L, \quad \mathbf{Q}_{hk}^0(u) = \mathbf{q}_{hk} + u\hat{\mathbf{c}}^*, \quad \mathbf{q}_{hk} = h\mathbf{a}^* + k\mathbf{b}^*, \quad (9)$$

where  $\hat{\mathbf{c}}^* = \mathbf{c}^*/|\mathbf{c}^*|$ .

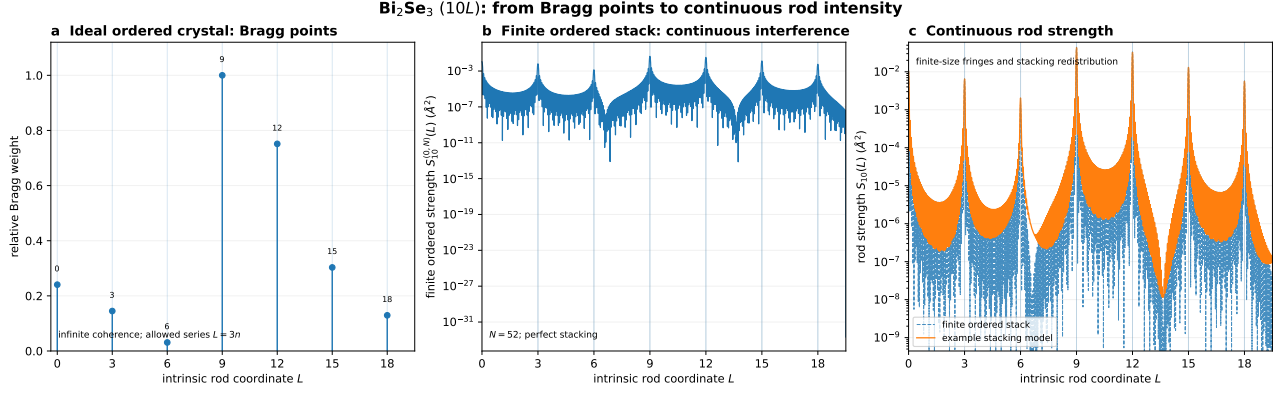


Figure 1: Construction of the Bi<sub>2</sub>Se<sub>3</sub> (10L) rod strength. (a) The infinite ordered crystal gives Bragg points at the allowed  $L = 3n$  positions. The stems represent delta functions with relative crystallographic weights. (b) A finite ordered stack with  $N = 52$  gives finite peak widths and finite-size fringes. (c) The stacking model redistributes intensity within that continuous profile.

## 2 Elastic reach defines the reciprocal-rod set

A rod can contribute only if an Ewald sphere can reach it for at least one incident wavelength. The shortest distance between the rod and the reciprocal origin is

$$Q_{R,hk} = |\mathbf{q}_{hk,\perp}|, \quad \mathbf{q}_{hk,\perp} = \mathbf{q}_{hk} - (\mathbf{q}_{hk} \cdot \hat{\mathbf{c}}^*) \hat{\mathbf{c}}^*. \quad (10)$$

Crystallite rotation preserves this distance.

An Ewald sphere of radius  $k$ , whose center is also a distance  $k$  from the reciprocal origin, can reach the rod when

$$Q_{R,hk} \leq 2k. \quad (11)$$

The shortest sampled wavelength has the largest wavevector,

$$k_{\max} = \frac{2\pi}{\lambda_{\min}}. \quad (12)$$

Every integer pair  $(h, k)$  satisfying

$$Q_{R,hk} \leq 2k_{\max} \quad (13)$$

is included in the complete rod set. For the Bi<sub>2</sub>Se<sub>3</sub> source distribution used here,  $\lambda_{\min} = 1.506747 \text{ \AA}$  and  $2k_{\max} = 8.340065 \text{ \AA}^{-1}$ . The complete rod set contains the  $(00L)$  rod and 84 nonzero  $(h, k)$  rods.

Each weighted incident-beam component applies the same reach condition using its own wavelength,

$$k_s = \frac{2\pi}{\lambda_s}, \quad Q_{R,hk} \leq 2k_s. \quad (14)$$

Thus the rods available for scattering depend slightly on wavelength. Of the 1000 incident-beam components, 991 can reach all 85 rods. The nine longest-wavelength components cannot reach the outer  $r = 21$  family and use 73 rods.

For the hexagonal in-plane lattice, rods with the same

$$r = h^2 + hk + k^2 \quad (15)$$

share the same powder-cylinder radius,

$$Q_R(r) = Q_R(1)\sqrt{r}, \quad Q_R(1) = 1.751194 \text{ \AA}^{-1}. \quad (16)$$

The value  $r$  labels a radial family. The number of rods is the number of distinct integer pairs  $(h, k)$  at that radius.

Table 1: Nonzero radial families in the  $\text{Bi}_2\text{Se}_3$  rod set. The final column is the number of distinct  $(h, k)$  rods at the listed radius.

| $r$ | $Q_R(r) \text{ (\AA}^{-1}\text{)}$ | number of rods |
|-----|------------------------------------|----------------|
| 1   | 1.751                              | 6              |
| 3   | 3.033                              | 6              |
| 4   | 3.502                              | 6              |
| 7   | 4.633                              | 12             |
| 9   | 5.254                              | 6              |
| 12  | 6.066                              | 6              |
| 13  | 6.314                              | 12             |
| 16  | 7.005                              | 6              |
| 19  | 7.633                              | 12             |
| 21  | 8.025                              | 12             |

The factor  $P_{hk}$  is the relative population assigned to one  $(h, k)$  rod. It allows unequal phase fractions or crystallographic-domain populations. Here  $P_{hk} = 1$  for every rod. For a compact figure, the rods at one radius are added as

$$S_{r,s}^\Sigma(L) = \sum_{\substack{(h,k) \text{ available at } \lambda_s \\ h^2+hk+k^2=r}} P_{hk} S_{hk}(L; \lambda_s). \quad (17)$$

This family sum is used only for display. The scattering calculation evaluates every  $(h, k)$  rod separately.

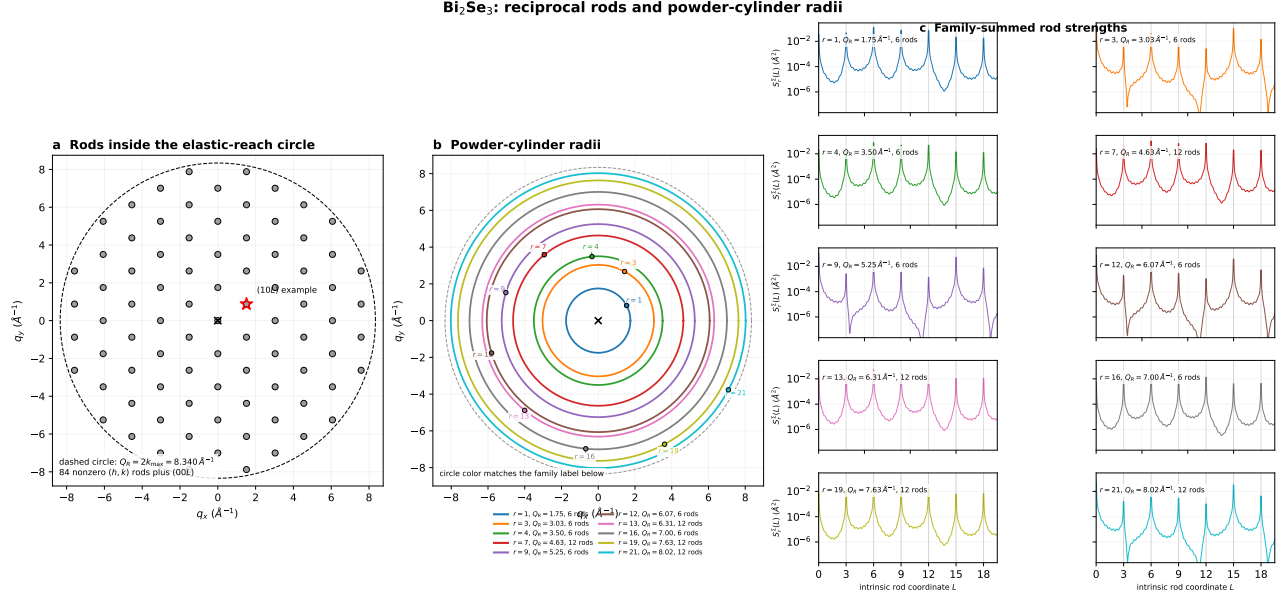


Figure 2: Reciprocal rods and their powder-cylinder radii. (a) Every integer rod inside the elastic-reach circle is included. The red star marks (10L) and the black cross marks (00L). (b) The ten powder-cylinder radii are color coded. Each ring is marked directly by  $r$ , and the matching legend gives  $Q_R$  and the number of rods. (c) Each small panel gives the family-summed strength  $S_r^\Sigma(L)$  for one discrete  $r$ .

### 3 Random in-plane orientation forms normalized powder cylinders

A two-dimensional powder contains crystallites with every in-plane azimuth. The rotation  $\mathbf{C}$  places the mean crystallite orientation in the sample frame and aligns the reciprocal stacking direction with the measured film normal,

$$\mathbf{C}\hat{\mathbf{c}}^* = \hat{\mathbf{n}}_{\text{film}}. \quad (18)$$

At zero mosaic tilt, rotation through crystallite azimuth  $\beta$  gives

$$\mathbf{Q}_{hk}^{\text{cyl}}(\beta, u) = \mathbf{C}R_{\hat{\mathbf{c}}^*}(\beta)\mathbf{Q}_{hk}^0(u), \quad 0 \leq \beta < 2\pi. \quad (19)$$

For fixed  $u$ , this traces a ring of radius  $Q_{R,hk}$ . Continuous  $u$ , or continuous  $L$ , stacks those rings into a cylinder.

A small rod segment  $du$  carries

$$dI_{hk,s}^{\text{rod}} = P_{hk}S_{hk}(L; \lambda_s) du. \quad (20)$$

Uniform crystallite azimuth assigns the fraction  $d\beta/(2\pi)$  to an angular interval  $d\beta$ ,

$$dI_{hk,s}^{\text{cyl}} = dI_{hk,s}^{\text{rod}} \frac{d\beta}{2\pi}. \quad (21)$$

The corresponding cylinder area is

$$dA_{\text{cyl},hk} = Q_{R,hk} d\beta du. \quad (22)$$

The intensity per reciprocal-space cylinder area is therefore

$$\rho_{\text{cyl},hk,s}(\beta, u) = \frac{P_{hk}S_{hk}(L; \lambda_s)}{2\pi Q_{R,hk}}. \quad (23)$$

Integrating around the circumference recovers the original rod segment,

$$\int_0^{2\pi} \rho_{\text{cyl},hk,s}(\beta, u) Q_{R,hk} d\beta = P_{hk}S_{hk}(L; \lambda_s). \quad (24)$$

The factor  $1/(2\pi Q_R)$  spreads the scattering in crystallite azimuth while preserving the intensity at each crystallographic  $L$ . Figure 3(a) identifies the ten discrete radial-family cylinders. The stepped display heights expose the nested surfaces, while every physical cylinder retains its full continuous  $L$  range. Panels (b) and (c) show the family densities and the numerical circumference integration.

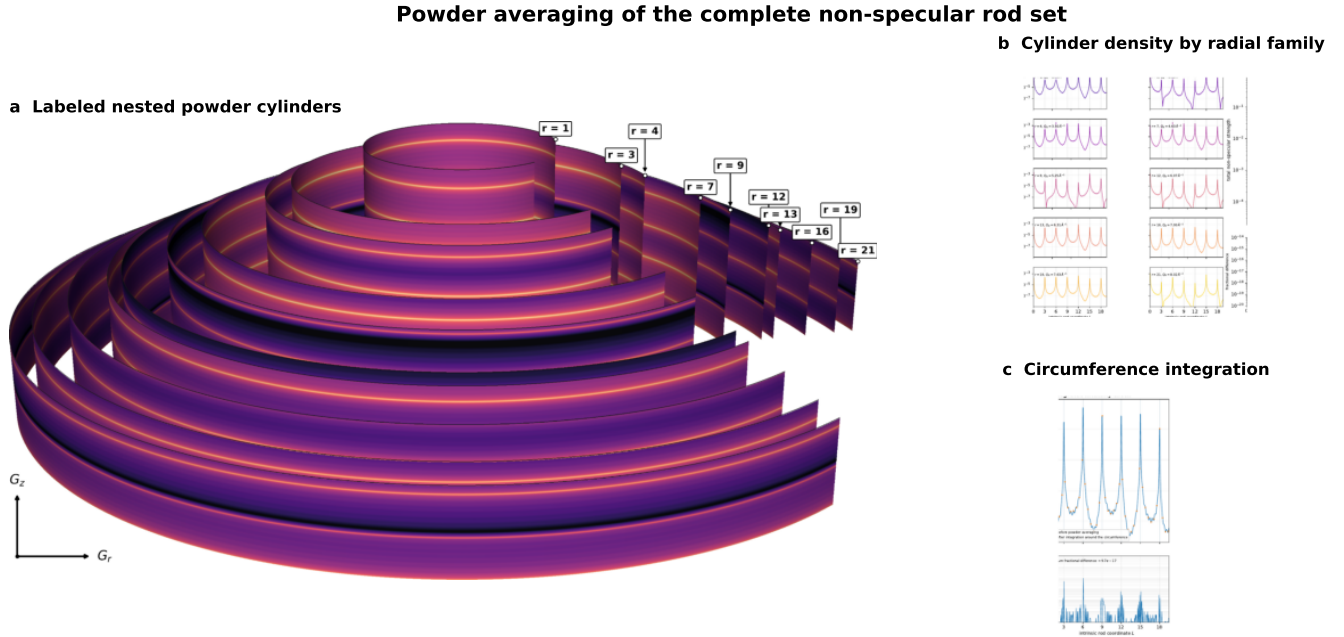


Figure 3: Powder averaging of the complete non-specular rod set. (a) The labeled cutaway identifies the cylinder associated with every discrete radial family  $r$ ; the stepped heights are used only to reveal the nested surfaces. (b) The cylinder density  $\rho_{\text{cyl},r}(L)$  for each family. (c) Circumference integration reproduces the summed rod intensity to numerical precision.

## 4 Mosaicity spreads the cylinders into reciprocal-space volume

Mosaicity rotates the complete crystallite, including the rod direction and its in-plane offset. Let  $\alpha$  be the tilt magnitude and  $\beta$  the azimuth of that tilt. If  $\hat{\mathbf{t}}$  is a reference in-plane tilt axis, then

$$Q_{hk}(\alpha, \beta, u) = \mathbf{C} \mathbf{R}_{\hat{\mathbf{c}}^*}(\beta) \mathbf{R}_{\hat{\mathbf{t}}}(\alpha) Q_{hk}^0(u). \quad (25)$$

The innermost rotation tilts the crystallite by  $\alpha$ . The second rotation sets the azimuth  $\beta$  of that tilt around the mean stacking direction. At  $\alpha = 0$ , the expression reduces to the powder cylinder.

The default  $\text{Bi}_2\text{Se}_3$  calculation uses a wrapped Gaussian tilt profile with width  $\sigma_\alpha = 1^\circ$ . Because tilt is an angular variable, the profile is repeated every  $2\pi$  so that probability is continuous across the angular boundary. The symmetric profile is then folded onto the nonnegative tilt magnitude  $0 \leq \alpha \leq \pi$ . The orientation model can also include a separately weighted Lorentzian tail, but its weight is zero in the default calculation shown here.

Let  $p_\alpha(\alpha)$  denote the resulting normalized tilt probability. Uniform crystallite azimuth gives

$$p_m(\alpha, \beta) = \frac{p_\alpha(\alpha)}{2\pi}, \quad \int_0^\pi \int_0^{2\pi} p_m(\alpha, \beta) d\beta d\alpha = 1. \quad (26)$$

For one rod in incident-beam component  $s$ , the intensity carried by  $d\alpha d\beta du$  is

$$dI_{hk,s}^{\text{pre}} = p_m(\alpha, \beta) P_{hk} S_{hk}(L; \lambda_s) d\alpha d\beta du. \quad (27)$$

Because  $p_m$  is normalized, integration over crystallite orientation recovers the original rod intensity.

The reciprocal-space volume generated by changes in  $\alpha$ ,  $\beta$ , and  $u$  is

$$dV_Q = J_{\text{vol},hk} d\alpha d\beta du, \quad J_{\text{vol},hk} = \left| \det \frac{\partial \mathbf{Q}_{hk}}{\partial(\alpha, \beta, u)} \right|. \quad (28)$$

When a broad range of crystallite orientations is compressed into a small reciprocal-space volume, the intensity per unit volume increases. Adding every rod and crystallite orientation that reaches the same reciprocal-space point gives

$$\rho_{Q,s}^{\neq 0}(\mathbf{Q}) = \sum_{\substack{\text{reachable } (h,k) \neq (0,0) \\ \text{scattering solutions at } \mathbf{Q}}} \frac{p_m(\alpha, \beta) P_{hk} S_{hk}(L; \lambda_s)}{J_{\text{vol},hk}}. \quad (29)$$

The intrinsic  $L$  in  $S_{hk}(L)$  remains the crystallite-frame rod coordinate. After rotation,

$$Q_z = \mathbf{Q} \cdot \hat{\mathbf{n}}_{\text{film}} \quad (30)$$

contains contributions from both the in-plane reciprocal vector and  $L\mathbf{c}^*$ . Thus  $Q_z c / (2\pi)$  is not generally equal to intrinsic  $L$  for a tilted crystallite.

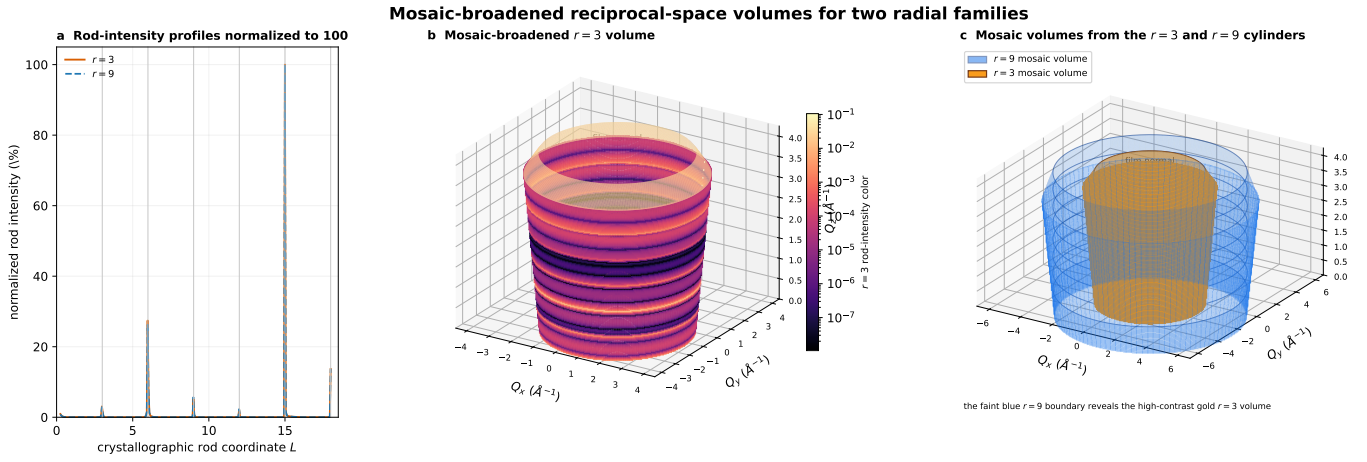


Figure 4: Mosaic broadening of selected off-specular radial families. (a) The  $r = 3$  and  $r = 9$  rod-intensity profiles are shown on a linear scale, with each profile normalized independently to a maximum of 100 for comparison of their  $L$  dependence. (b) The boundary of the reciprocal-space volume formed from the  $r = 3$  powder cylinder by the full orientation range  $0 \leq \alpha \leq 10^\circ$  and  $0 \leq \beta < 2\pi$ ; the interior contains all smaller tilt magnitudes. The surface color follows the  $r = 3$  rod intensity along  $L$ . (c) Applying the same mosaic construction to the  $r = 3$  and  $r = 9$  cylinders produces two nested reciprocal-space volumes. The outer  $r = 9$  boundary is rendered as a faint blue shell, while the inner  $r = 3$  volume is shown in high-contrast gold so that both families remain visible. The colors identify the families and do not compare their absolute intensities.

## 5 The $(00L)$ family remains axial until mosaic tilt

The  $r = 0$  family contains only the  $(00L)$  rod,

$$Q_{00}^0(L) = Lc^*, \quad Q_R = 0. \quad (31)$$

The same ideal-to-finite construction produces a continuous  $S_{00}(L)$ . The lateral registry phase is unity for  $h = k = 0$ , so the finite ordered and example stacking-model profiles nearly coincide in the present registry-transition model.

At zero tilt, changing  $\beta$  leaves the  $00L$  vector on the specular axis. Once  $\alpha > 0$ , a fixed crystallographic  $L$  moves to

$$Q_R = |Lc^*| \sin \alpha, \quad Q_z = |Lc^*| \cos \alpha, \quad (32)$$

while  $\beta$  distributes the tilted vector around the film normal. Figure 5(c) shows the full  $0 \leq \alpha \leq 10^\circ$  broadening of the  $L = 9$  and  $L = 18$  points. Figure 5(d) integrates continuous  $L$  over the same tilt range to give the specular reciprocal-space volume. Figure 5(e) shows the closed boundary of that full volume: the side corresponds to  $\alpha = 10^\circ$  and the top cap contains all tilts from  $0$  to  $10^\circ$  at the largest displayed  $L$ .

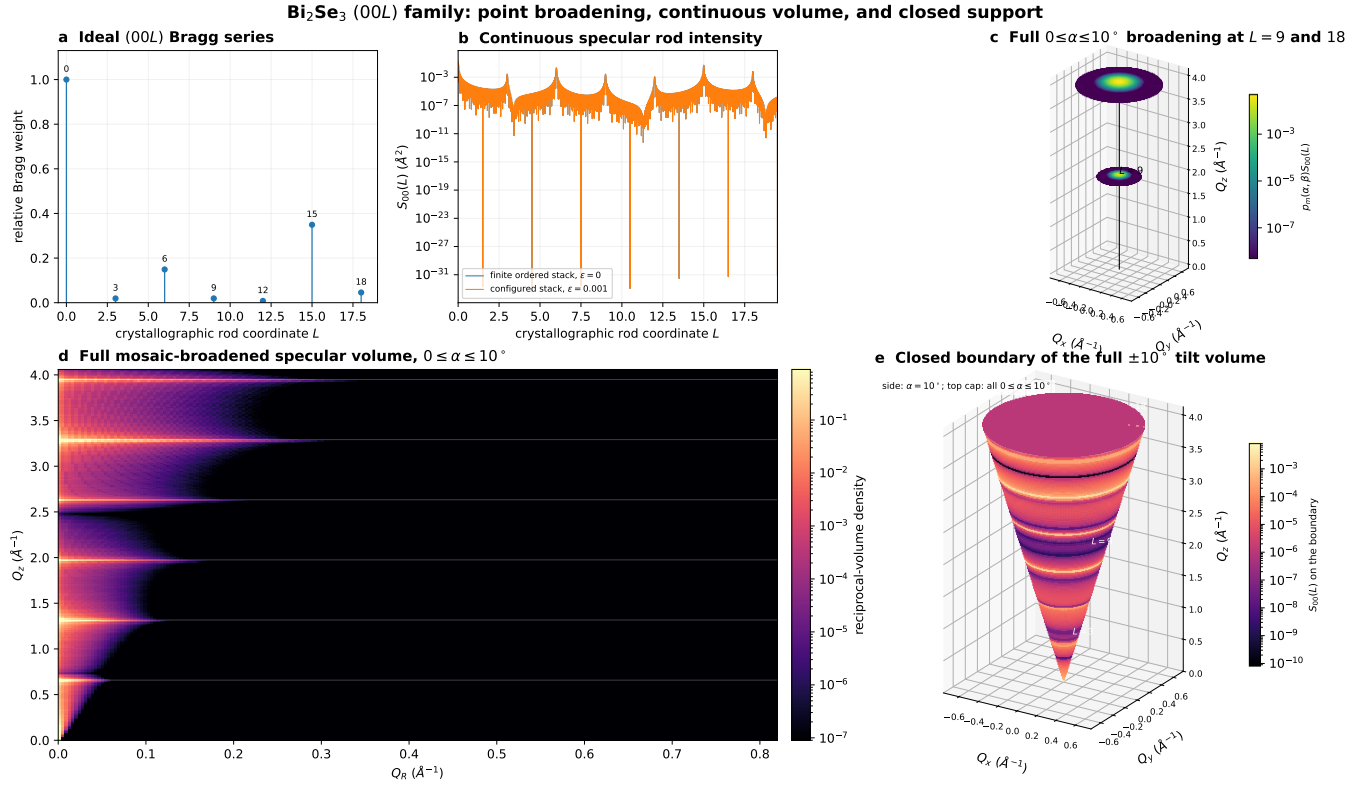


Figure 5: Construction of the Bi<sub>2</sub>Se<sub>3</sub> (00L) family. (a) The infinite ordered limit gives the  $L = 3n$  Bragg series. (b) The finite ordered and stacking-model profiles nearly overlap for the present  $r = 0$  registry phase. (c) The complete  $0 \leq \alpha \leq 10^\circ$  orientation range broadens the individual  $L = 9$  and  $L = 18$  axis points into finite-radius surfaces. (d) Continuous  $L$  and the full tilt range produce the mosaic-broadened specular volume. (e) The boundary of that volume is a closed cone: its side is the  $\alpha = 10^\circ$  surface and its top cap contains the full range  $0 \leq \alpha \leq 10^\circ$  at the largest displayed  $L$ .

## 6 Elastic scattering restricts the reciprocal density to the Ewald sphere

### 6.1 Off-specular rods form detector-visible Ewald patches

For one incident-beam component  $s$ , the internal incident wavevector is  $\mathbf{k}_{i,s}$  and

$$K_s = |\mathbf{k}_{i,s}|. \quad (33)$$

Elastic scattering inside the film requires

$$|\mathbf{k}_{f,s}| = K_s, \quad \mathbf{Q} = \mathbf{k}_{f,s} - \mathbf{k}_{i,s}, \quad (34)$$

so the allowed reciprocal vectors lie on

$$|\mathbf{k}_{i,s} + \mathbf{Q}| = K_s. \quad (35)$$

This Ewald sphere has radius  $K_s$ , center  $-\mathbf{k}_{i,s}$ , and the reciprocal origin on its surface.

The preceding reciprocal-space construction supplies a continuous density before this elastic restriction. The Ewald condition retains only the rod and crystallite orientations that intersect the sphere for the selected incident-beam component. The following pages first describe this restriction with the analytic rod-sphere geometry and then show the equivalent detector-coordinate evaluation used by the production calculation.

For a fixed off-specular rod and crystallite orientation, the reciprocal rod is a straight line,

$$\mathbf{Q}(u) = \mathbf{q}_0 + u\hat{\mathbf{d}}, \quad (36)$$

where  $\mathbf{q}_0$  is the point at  $u = 0$  and  $\hat{\mathbf{d}}$  is the rotated rod direction. Defining

$$\mathbf{a} = \mathbf{k}_{i,s} + \mathbf{q}_0, \quad \mathbf{a}_\perp = \mathbf{a} - (\mathbf{a} \cdot \hat{\mathbf{d}})\hat{\mathbf{d}}, \quad (37)$$

the algebraic intersections with the internal Ewald sphere are

$$u_j = -\mathbf{a} \cdot \hat{\mathbf{d}} \pm \sqrt{K_s^2 - |\mathbf{a}_\perp|^2}, \quad L_j = \frac{u_j}{|\mathbf{c}^*|}. \quad (38)$$

A negative value under the square root means that the oriented rod misses the sphere, zero gives tangency, and a positive value gives two algebraic intersections.

These roots are used to construct a forward Ewald-patch picture and to validate the branch geometry. They are not evaluated as a separate line-intersection step when the detector image is calculated. In the detector calculation, a detector coordinate fixes the outgoing ray and therefore fixes a reciprocal-space point on the Ewald sphere,

$$(c, r_d) \longrightarrow \mathbf{k}_{f,s}^{\text{air}} \longrightarrow \mathbf{k}_{f,s}^{\text{film}} \longrightarrow \mathbf{Q}_s \longrightarrow \{h, k, j, \alpha, \beta, L\}_{\text{compatible}}. \quad (39)$$

The inverse reciprocal map then returns every rod, crystallite orientation, branch, and crystallographic  $L$  that contributes to that detector point. Figure 6 shows the complete intrinsic Ewald intensity for the nominal  $5^\circ$  incidence state. Figure 7 then isolates the subset that reaches the active detector.

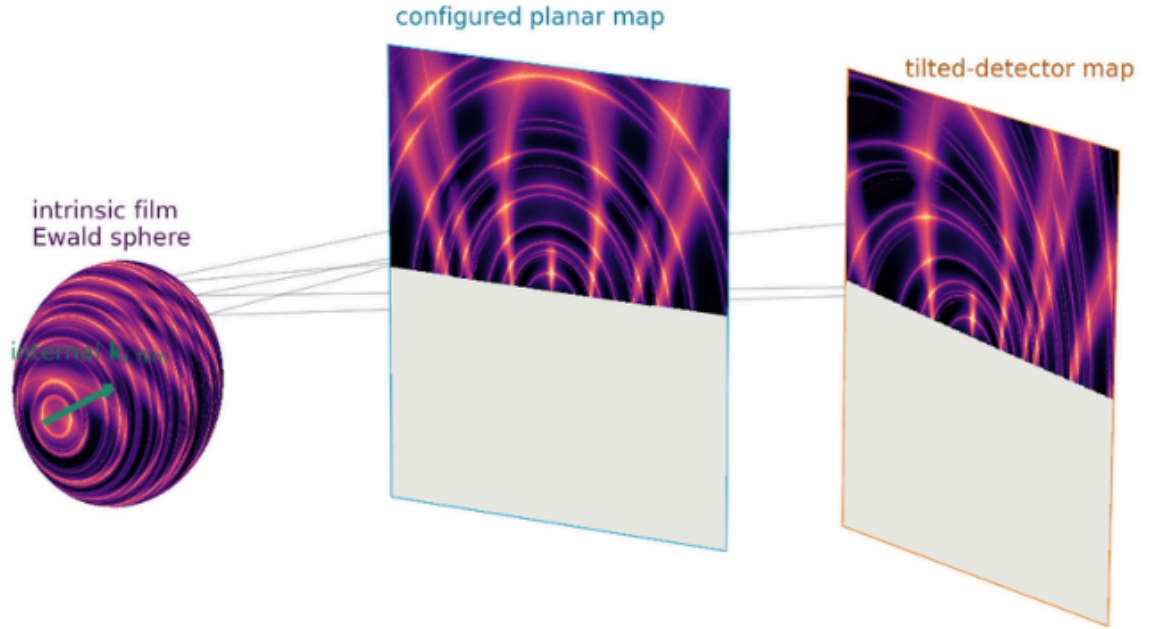
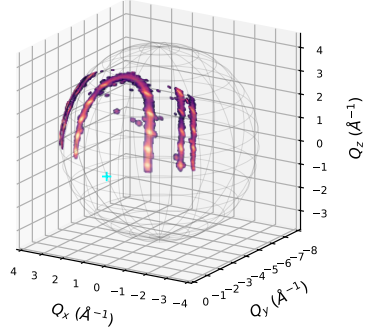


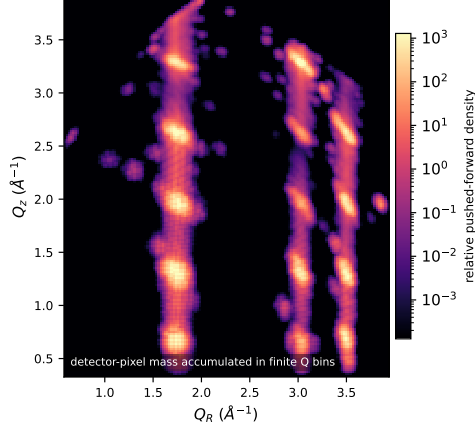
Figure 6: Complete intrinsic Ewald intensity and detector mappings for the nominal  $\text{Bi}_2\text{Se}_3$  incidence angle of  $5^\circ$ . The full internal-film Ewald sphere is shown at left, followed by the configured planar map and the tilted-detector map after exit refraction. Figure 7 isolates the active-detector subset.

# Off-specular detector-visible Ewald support, nominal incidence $5^\circ$

a Ewald points: detector-side view



b Area-binned  $(Q_R, Q_z)$  density



c Detector density

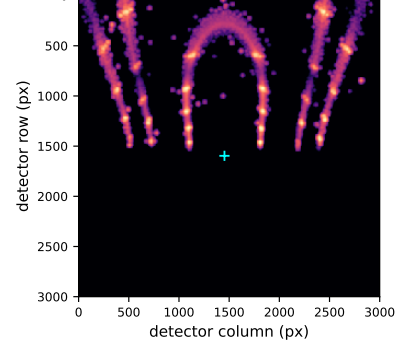


Figure 7: Detector-visible off-specular Ewald selection for the nominal  $\text{Bi}_2\text{Se}_3$  incidence angle of  $5^\circ$ . (a) The view is rotated toward the detector side so the detector-visible reciprocal-space points lie on the near side of the sphere; the cyan cross marks  $\mathbf{Q} = 0$ . (b) The detector-pixel intensity is accumulated in finite  $(Q_R, Q_z)$  bins and divided by the bin area. This area-binned pushforward removes the visual gaps and artificial under-representation near smaller  $|\mathbf{Q}|$  that occur when irregularly spaced mapped points are plotted individually. (c) The detector density that supplies panels (a) and (b). This figure isolates the detector-visible subset of the complete sphere shown in Fig. 6.

## 6.2 The specular family retains the nonzero kinematic branch

The nonzero kinematic branch is treated separately because the  $00L$  rod passes through the reciprocal origin. For one crystallite orientation, write the rod as

$$\mathbf{Q}(u) = u\hat{\mathbf{d}}. \quad (40)$$

The elastic condition is

$$|\mathbf{k}_i + u\hat{\mathbf{d}}|^2 = |\mathbf{k}_i|^2. \quad (41)$$

Expanding the left-hand side gives

$$u(2\mathbf{k}_i \cdot \hat{\mathbf{d}} + u) = 0. \quad (42)$$

The first solution,  $u = 0$ , is the direct-beam point  $\mathbf{Q} = 0$ . The second solution is the nonzero kinematic  $00L$  branch,

$$u_{00}^{\text{kin}} = -2\mathbf{k}_i \cdot \hat{\mathbf{d}}. \quad (43)$$

If the rod direction is represented by a non-unit vector  $\mathbf{d}$ , the same derivation gives

$$u_{00}^{\text{kin}} = -2 \frac{\mathbf{k}_i \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{d}}. \quad (44)$$

This derivation explains why the nonzero solution is used: the  $u = 0$  root identifies the direct beam, while the second root is the kinematic scattering point on the  $00L$  rod.

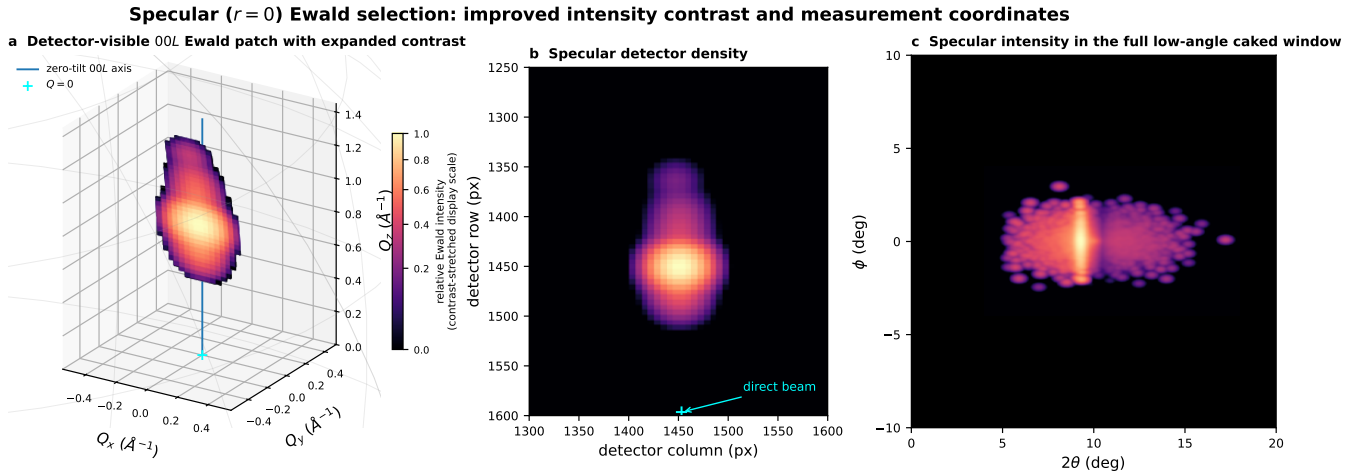


Figure 8: Specular ( $r = 0$ ) Ewald selection for the nominal  $\text{Bi}_2\text{Se}_3$  incidence angle of  $5^\circ$ . (a) A local view centered on the detector-visible  $00L$  Ewald patch, with the zero-tilt  $00L$  axis and direct-beam point  $\mathbf{Q} = 0$  retained for reference. The color scale is reconstructed from the rendered detector field and contrast-stretched over its populated range so that the internal intensity variation across the patch is visible; it is a relative display scale rather than an additional physical normalization. (b) The corresponding specular detector density; the cyan cross marks the direct-beam coordinate. (c) The same specular intensity in the caked representation over the wider low-angle window  $0^\circ \leq 2\theta \leq 20^\circ$  and  $-10^\circ \leq \phi \leq 10^\circ$ . The calculated feature occupies the central part of this displayed angular region.

### 6.3 The crossing factor cancels in the physical Ewald-surface density

The Ewald condition restricts a continuous rod coordinate to its regular roots. Let

$$g(u) = |\mathbf{k}_i + \mathbf{Q}(u)| - K. \quad (45)$$

The one-dimensional delta-function identity

$$\int f(u) \delta[g(u)] du = \sum_j \frac{f(u_j)}{|g'(u_j)|}, \quad g(u_j) = 0, \quad (46)$$

shows why a line-sphere crossing factor appears. Since  $d\mathbf{Q}/du = \hat{\mathbf{d}}$  and  $\mathbf{k}_f = \mathbf{k}_i + \mathbf{Q}(u_j)$ ,

$$g'(u_j) = \frac{\mathbf{k}_f \cdot \hat{\mathbf{d}}}{K}, \quad J_E = \frac{1}{|g'(u_j)|} = \frac{K}{|\mathbf{k}_f \cdot \hat{\mathbf{d}}|}. \quad (47)$$

Thus  $J_E$  is the Jacobian produced by restricting the continuous rod measure to the Ewald sphere. In the forward orientation coordinates, the selected intensity is

$$dI_{hk,j,s}^E = p_m(\alpha, \beta) P_{hk} S_{hk}(L_j; \lambda_s) J_{E,hk,j,s} d\alpha d\beta. \quad (48)$$

The same orientation interval spans the physical Ewald-surface area

$$dA_Q = J_E J_{\text{vol,hk}} d\alpha d\beta, \quad J_{\text{vol,hk}} = \left| \det \frac{\partial \mathbf{Q}}{\partial(\alpha, \beta, u)} \right|. \quad (49)$$

Dividing the selected intensity by this area gives

$$\rho_{Q,\text{surf},s}(\mathbf{Q}) = \sum_{\substack{\text{rods and branches} \\ \text{compatible with } \mathbf{Q}}} \frac{p_m(\alpha, \beta) P_{hk} S_{hk}(L_j; \lambda_s)}{J_{\text{vol,hk}}}. \quad (50)$$

Because  $dA_Q = K_s^2 d\Omega_{\text{film}}$ , the density per internal-film outgoing solid angle is

$$\rho_{\Omega,\text{film},s} = \sum_{\substack{\text{rods and branches} \\ \text{compatible with } \mathbf{Q}}} \frac{p_m(\alpha, \beta) P_{hk} S_{hk}(L_j; \lambda_s) K_s^2}{J_{\text{vol,hk}}}. \quad (51)$$

## 7 Detector geometry maps Ewald intensity into detector, angle, and $Q$ coordinates

Each weighted incident-beam component  $s$  supplies a wavelength  $\lambda_s$ , incident direction, sample intersection, statistical weight  $w_s$ , and internal incident vector  $\mathbf{k}_{i,s}^{\text{film}}$ . A detector coordinate  $(c, r_d)$  identifies a physical detector point and its outgoing air ray. Exit refraction gives the internal outgoing vector  $\mathbf{k}_{f,s}^{\text{film}}$ , and

$$\mathbf{Q}_s(c, r_d) = \mathbf{k}_{f,s}^{\text{film}}(c, r_d) - \mathbf{k}_{i,s}^{\text{film}}. \quad (52)$$

For this  $\mathbf{Q}_s$ , the inverse reciprocal map determines every compatible rod, branch, crystallite orientation, and crystallographic  $L$ . The rod intensity is evaluated at each recovered  $L$ .

For one incident-beam component, the detector intensity density is

$$D_s(c, r_d) = \sum_{\substack{\text{compatible } (h,k),j \\ \alpha, \beta, L}} \frac{p_m(\alpha, \beta) P_{hk} S_{hk}(L; \lambda_s)}{J_{\text{vol,hk}}} J_{Q \leftarrow cr,s} W_{\text{opt},s}, \quad (53)$$

where

$$J_{Q \leftarrow cr, s} = \left| \frac{\partial \mathbf{Q}_s}{\partial c} \times \frac{\partial \mathbf{Q}_s}{\partial r_d} \right| \quad (54)$$

converts detector-coordinate area into internal Ewald-surface area. It contains detector solid angle, detector pose and distance, source-point parallax, and the area change from exit refraction. The optical weight is

$$W_{\text{opt}, s} = |t_{\text{in}, s} t_{\text{out}, s}|^2 A_{\text{depth}, s}. \quad (55)$$

The source-averaged detector density and finite-pixel intensity are

$$D(c, r_d) = \sum_s w_s D_s(c, r_d), \quad I_{pq} = \int_{\text{pixel } pq} D(c, r_d) \, dc \, dr_d. \quad (56)$$

The same detector-visible intensity can be written in caked coordinates  $(2\theta, \phi)$ ,

$$S_{2\theta\phi}(2\theta, \phi) = D(c, r_d) \left| \frac{\partial(c, r_d)}{\partial(2\theta, \phi)} \right|, \quad (57)$$

or in source-resolved internal reciprocal-space coordinates,

$$Q_R = \sqrt{Q_x^2 + Q_y^2}, \quad Q_z = \mathbf{Q} \cdot \hat{\mathbf{n}}_{\text{film}}. \quad (58)$$

The detector-to- $\mathbf{Q}$  transformation is performed separately for each incident-beam component before the intensities are added, because a fixed detector point corresponds to a different  $\mathbf{Q}_s$  for different wavelengths and incident directions.

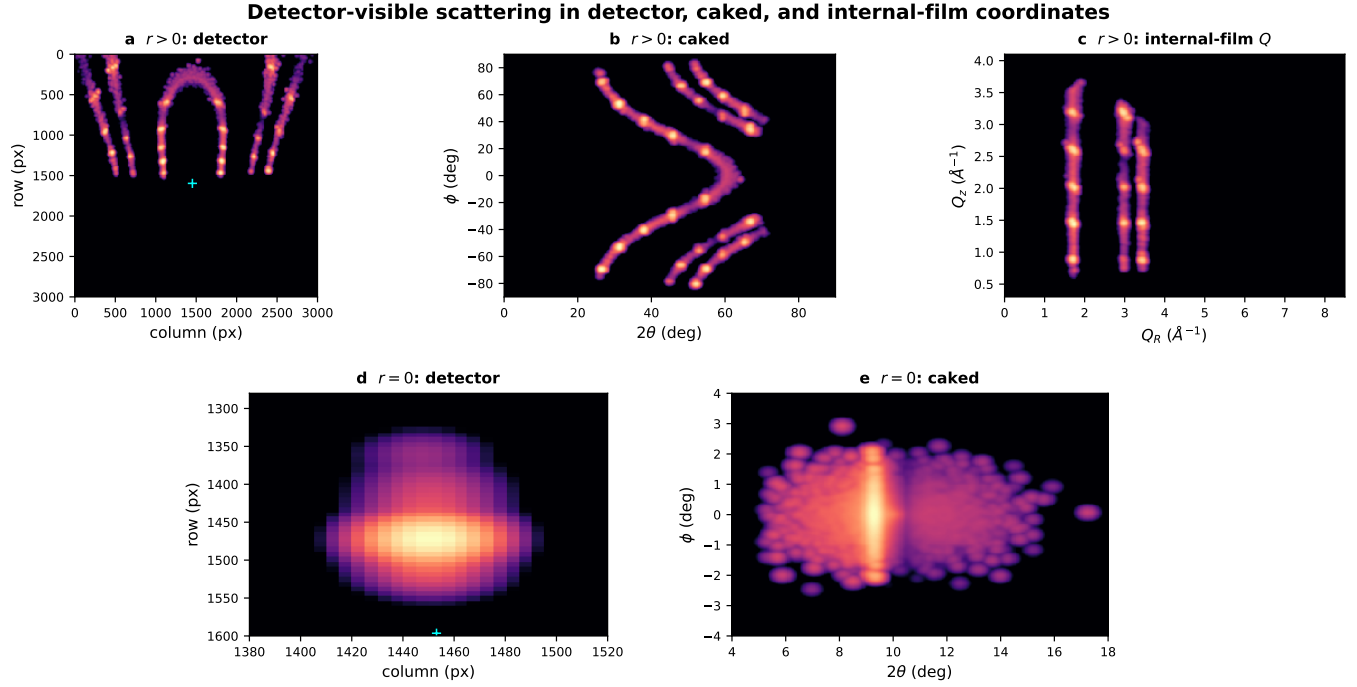


Figure 9: Detector-visible scattering in measurement coordinates. Panels (a)–(c) show the summed off-specular ( $r > 0$ ) intensity in detector, caked, and internal-film ( $Q_R, Q_z$ ) coordinates. Panels (d) and (e) show the specular ( $r = 0$ ) intensity in detector and caked coordinates. The caked azimuth  $\phi$  describes the outgoing ray, whereas  $\beta$  denotes crystallite azimuth.